Multi Tier Computing
DCE: The Enterprise NOS
(Unit 4)

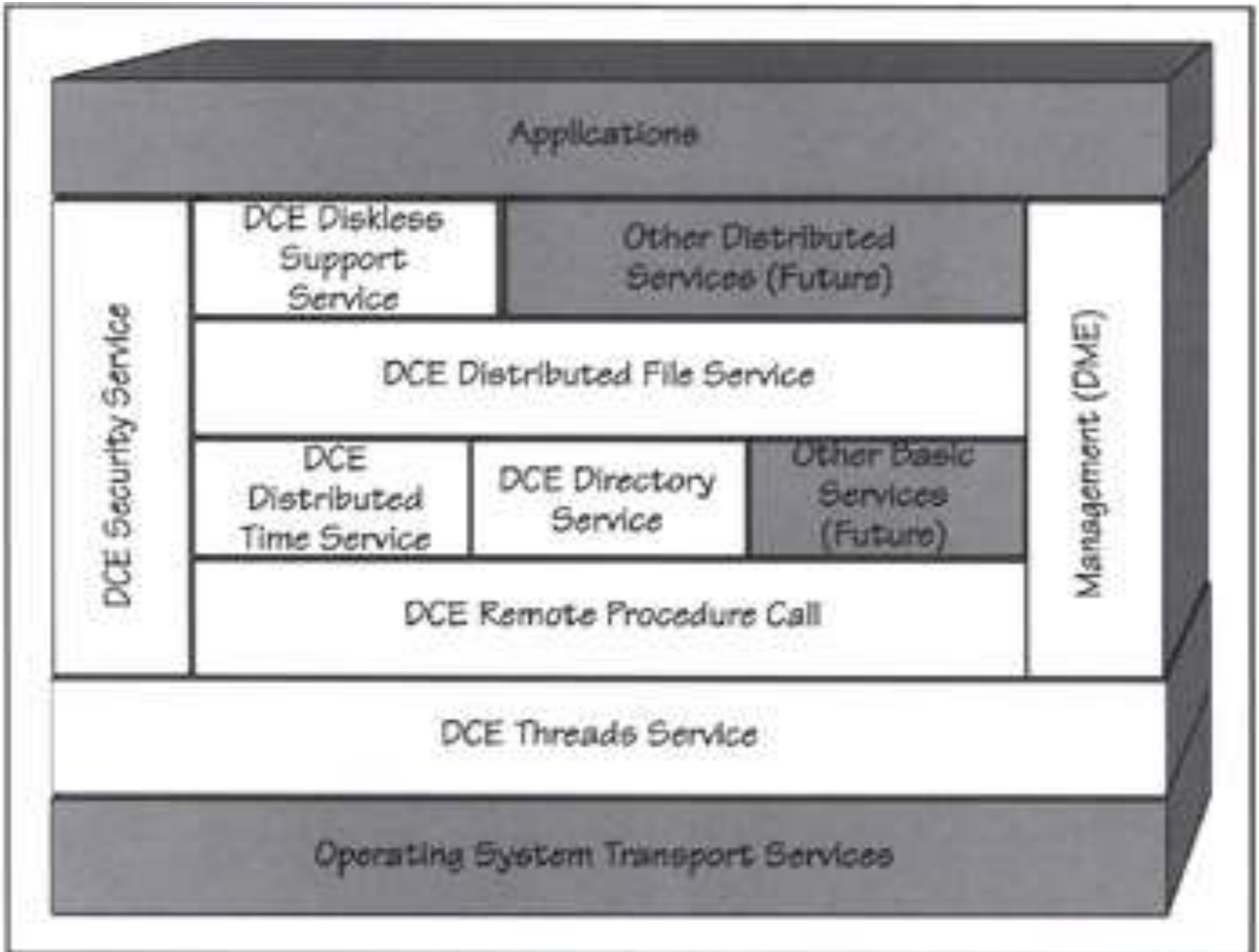
DISTRIBUTED COMPUTING ENVIRONMENT

The Enterprise NOS, The Internet as a NOS, MULTI-TIER COMPUTING: Overview –Benefits –Disadvantages-Components –Tier separations and Interaction.

THIN CLIENT COMPUTING: Introduction to computing models – Comparison – Components-environments

DCE

- The Distributed Computing Enviroment (DCE) from the **Open Software Foundation** (OSF)
- creates an open Enterprise/NOS environment that **spans multiple architectures, protocols, and operating systems**.
- DCE provides key distributed technologies, including a **remote procedure call**, a **distributed naming service**, a **time synchronization service**, **a distributed file system**, a **network security service**, and a **threads package**


Applications
DCE Diskless Support Service
Other Distributed Services (Future)
DCE Security Service
DCE Distributed File Service
Management (DME)
DCE Distributed Time Service
DCE Directory Service
Other Basic Services (Future)
DCE Remote Procedure Call
DCE Threads Service
Operating System Transport Services

- DCE allows a client to interoperate with one or more server processes on other computing platforms, even when they are from different vendors with different operating systems.

- DCE provides an integrated approach to security, naming, and interprocess communications.

DCE RPC

- The DCE RPC initially came from HP
- DCE **provides** an ***Interface Definition Language (IDL)*** and **compiler** that facilitate the creation of RPCs.

- The **IDL compiler creates portable C code stubs** for both the client and server sides of an application.
- The stubs are compiled and linked to the RPC run-time library, which is responsible for
  - **finding servers** in a distributed system,
  - performing the **message exchanges**,
  - **packing and unpacking** message parameters, and
  - processing any **errors** that occur.

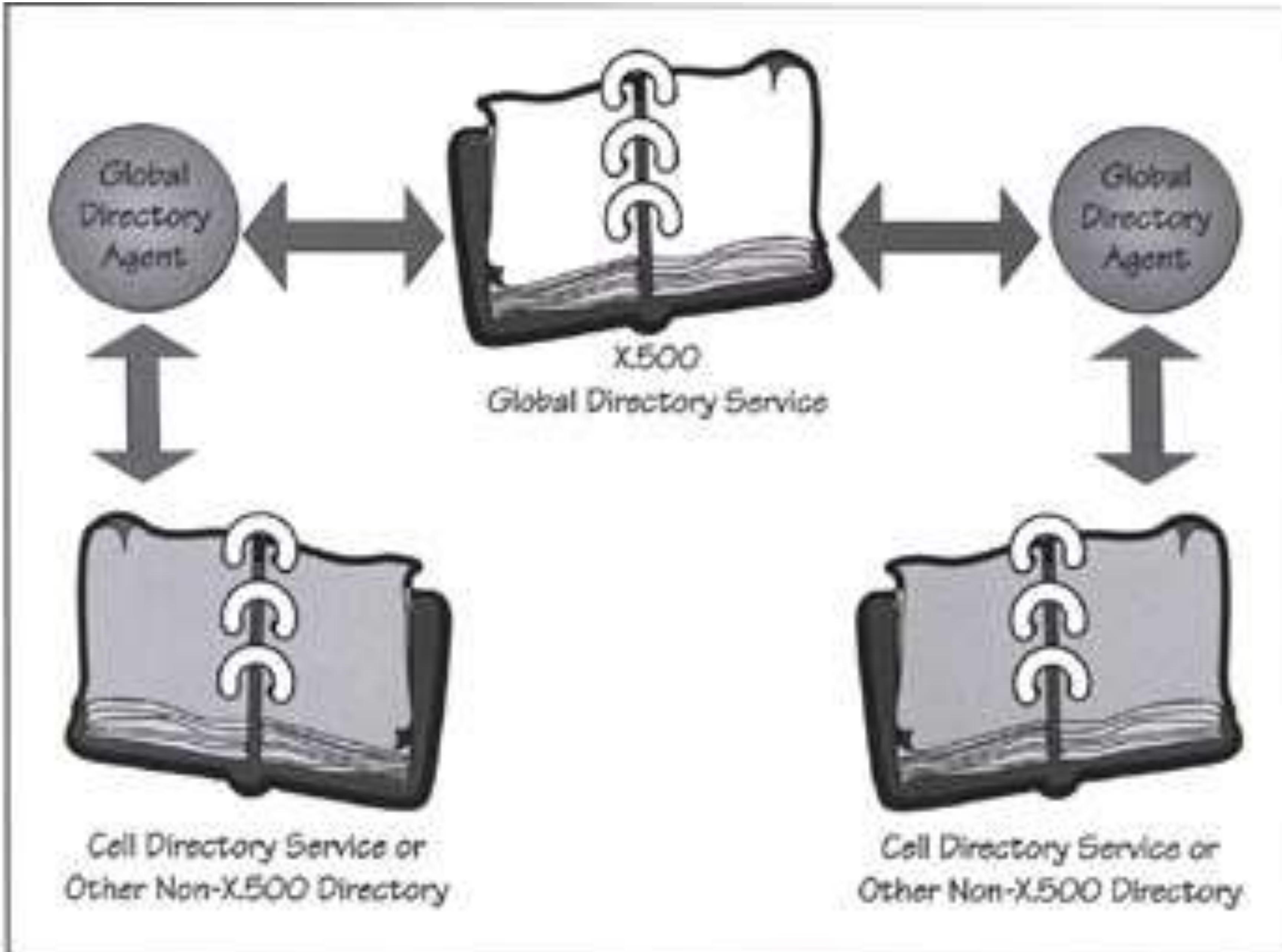
DCE RPC

- DCE RPC **can be integrated with the DCE security** and naming services.
  - This integration makes it possible **to authenticate each procedure call** and to dynamically locate servers at run time.
- **Servers can concurrently service RPCs using threads.**
- The RPC mechanism provides protocol and network independence.

DCE: Distributed Naming Services

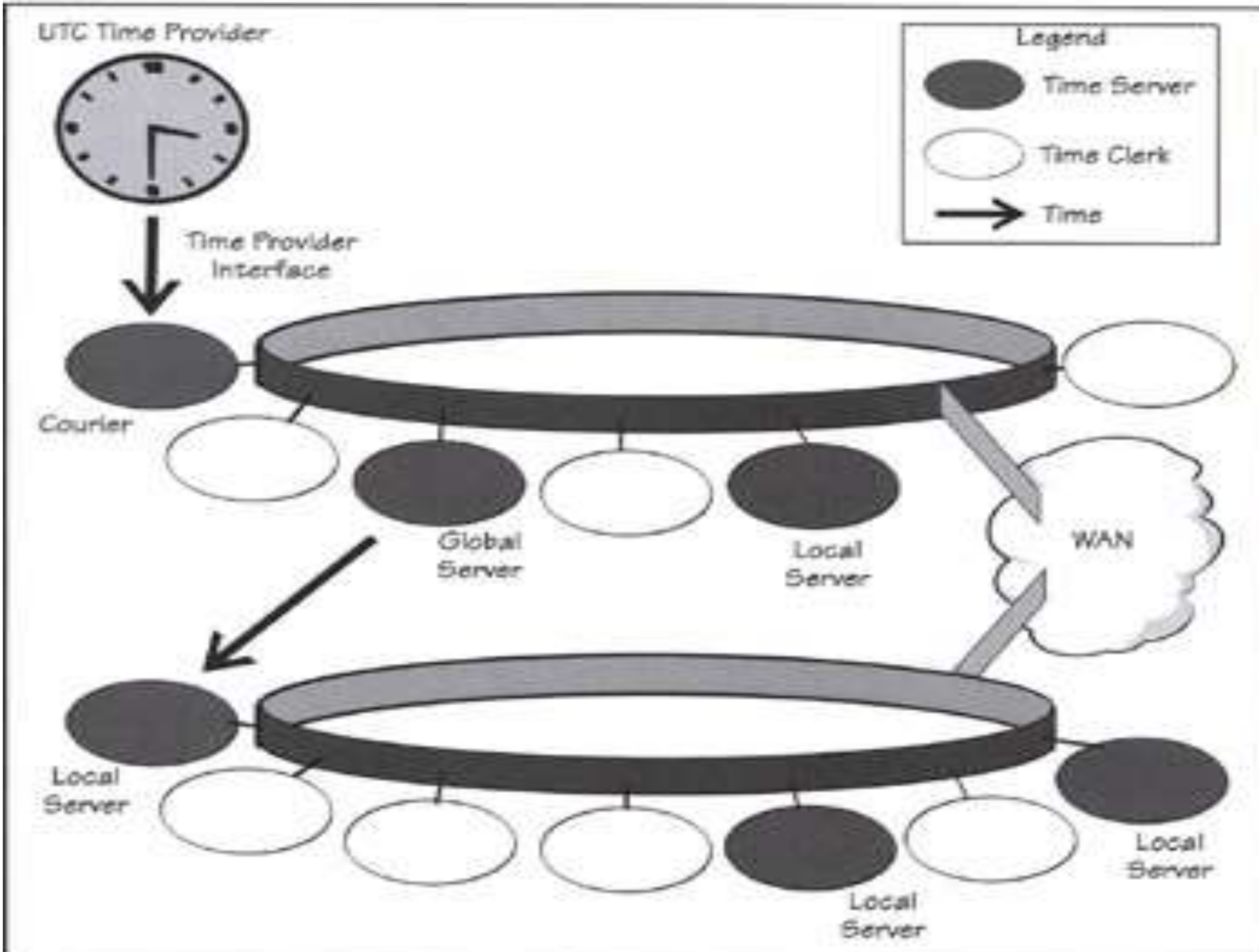
- Based on X.500
- The DCE naming services
  - allow **resources** such as programs, servers, files, disks, or print queues **to be identified by user-oriented names**
  - in a **special-purpose distributed database** that describes the objects of interest
  - Object names are **independent of their location** on the network.

- DCE divides the distributed environment into administrative units (or domains) called **cells**.
  - **DCE cell is a combination of client and server workstations**.
  - The cell's **domain is defined** by the customer.
  - A cell usually **consists of the set of machines** used by one or more groups working on related tasks.
  - The cell size is dictated only by how easy it is to administer.
  - At a minimum, **a DCE cell must have one cell directory server and one security server**.


Global Directory Agent
X.500
Global Directory Service
Global Directory Agent
Cell Directory Service or Other Non-X.500 Directory
Cell Directory Service or Other Non-X.500 Directory

DCE: Distributed Time Service

- a mechanism for synchronizing each computer in the network to a recognized time standard.
- The DCE Time Service provides APIs for manipulating timestamps and for obtaining universal time from public sources such as the Traconex/PSTI radio clock.
- DCE requires at least three Time Servers; one (or more) must be connected to an External Time Provider
- The Time Servers periodically query one another to adjust their clocks.
- The External Time Provider may be a hardware device that receives time from a radio or a telephone source.
- DCE uses the UTC standard, which keeps track of time elapsed since the beginning of the Gregorian calendar—October 15, 1582


UTC Time Provider
Legend
Time Server
Time Clerk
Time
Time Provider
Interface
Courier
Global
Server
Local
Server
WAN
Local
Server
Local
Server
Local
Server

DCE: Distributed Security Services

- OSF adopted MIT's Kerberos authentication system and enhanced it with some HP security features.
  - based on total mutual distrust
  - The Kerberos monster consists of three "heads," all residing on the same secured server: the Authentication Server, Security Database, and Privilege Server
- DCE's network security services provide authentication, authorization, and user account management.
- The DCE security server is a physically secured server that stores security-related information such as names and associated passwords.
  - The security agent works with the authenticated RPC to provide secure access to all the DCE services

- The **RPC mechanism** hides all the complexity of the security system from the user.
  - It obtains the tickets, provides encryption when needed,
  - and performs authenticated checksums if the policy requires it.

- CE supports authorization through Access Control Lists (ACLs).
  - Each DCE implementation that uses ACLs must implement an ACL manager that controls access to services and resources managed by it

Distributed File System (DFS)

- The DCE Distributed File System (DFS) provides a uniform namespace, file location transparency, and high availability.
- DFS is log-based and thus offers the advantage of a fast restart and recovery after a server crash.
- Files and directories can be replicated (invisibly) on multiple servers for high availability.
- A cache-consistency protocol allows a file to be changed in a cache.
- The changes are automatically propagated to all other caches where the file is used, as well as on the disk that owns the file
- DFS provides a single-image file system that can be distributed across a group of file servers.

Threads

- OSF chose the Concert Multithread Architecture (CMA) from Digital
- This portable thread package runs in the user space and includes small wrapper routines to translate calls to a native kernel-based thread package (like NT or Mach threads).
- Threads are an essential component of client/server applications and are used by the other DCE components

Enterprise NOS (Network Operating System)

- An Enterprise Network Operating System (NOS) is a system that manages network resources, users, security, and services across an organization.

Examples

- Windows Server
- Linux Server (Ubuntu Server, Red Hat)
- UNIX Systems

Features of Enterprise NOS

- Centralized user management
- Resource sharing (files, printers)
- Security & access control
- Network monitoring
- Backup and recovery
- Scalability for large organizations

Functions of Enterprise NOS

- Authentication & authorization
- Directory services (LDAP/Active Directory)
- Distributed file systems
- Communication services
- Remote administration
- Load balancing

Internet as a Network Operating System (NOS)

The Internet itself functions as a global NOS by providing shared services across distributed systems.

Provides:

- Communication infrastructure
- Resource access
- Distributed applications
- Global connectivity

Services Provided by Internet as NOS

- Email services
- Web services (HTTP)
- Cloud computing
- Remote login (SSH)
- File transfer (FTP)
- DNS name resolution

The Internet as a NOS

SSL (now TLS)

- As the name implies, **Secure Sockets Layer (SSL) is a secured socket connection.**
- Today, **all commercial browsers and Web servers** support SSL; it is the standard for authentication and encryption between Web browsers and servers.
- SSL also verifies that the content of the message hasn't been altered.
- SSL implements a security-enhanced version of sockets that provides transaction security at the transport level.
- SSL provides an unobtrusive security layer between the TCP/IP transport and sockets.
- In other words, SSL provides a secured communications link without involving the applications that invoke it; for example, the existence of SSL is transparent to e-mail or HTTP.
- HTTP servers that implement SSL must run on socket port 443 instead of the standard 80; it has become a common practice for firewall vendors to leave this port open.

The SSL protocol provides:

- 1) private client/server interactions using encryption,
- 2) server authentication, and
- 3) reliable client/server exchanges via message integrity checks that detect tampering.
- When a client and server first start communicating, they agree on an SSL protocol version, select the cryptographic algorithms, optionally authenticate each other, and use public-key encryption techniques to generate shared secrets.
- The SSL handshake protocol must first be completed before an application can transmit or receive its first byte of data.
- The handshake protocol is like a dance that allows the client and server to authenticate each other and to negotiate an encryption algorithm.

- Typically, the browser needs to ensure that the server is using a valid certificate.
- This is how you protect yourself against rogue server impersonators.
- Servers can also use SSL to verify a client's credentials (or certificate).
- An SSL session may include multiple secure connections;
- in addition, you can have multiple simultaneous sessions.
- The encryption established between a client and a server remains valid over the multiple connections.
- SSL uses public key authentication and encryption technology developed by RSA Data Security

S-HTTP

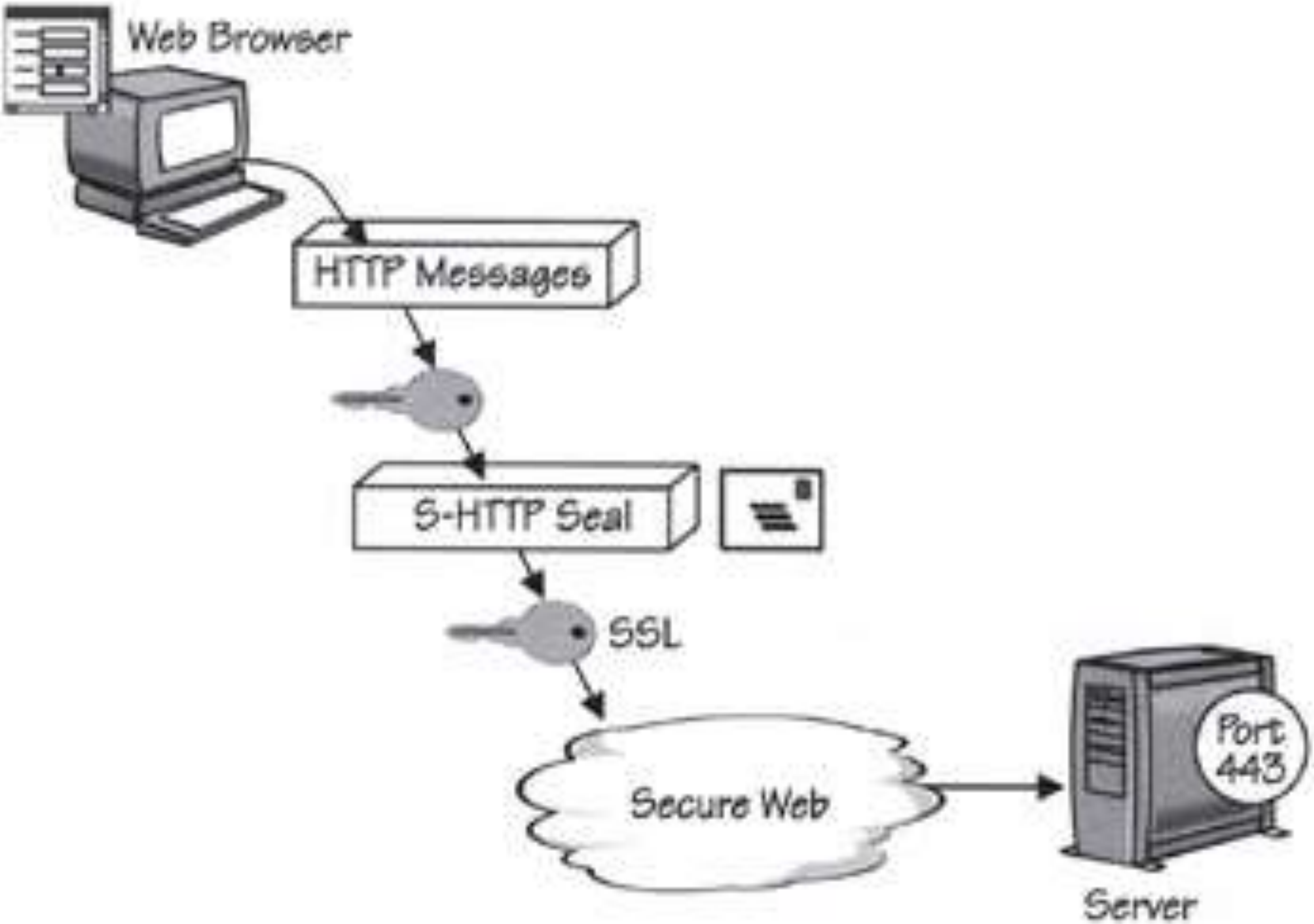
- S-HTTP is a security-enhanced variant of the Internet's Hypertext Transport Protocol (HTTP) that was developed by EIT.

- S-HTTP adds application-level encryption and security on top of ordinary sockets-based communications. The client and server communicate over an ordinary HTTP session and then negotiate their security requirements; they use a MIME-like protocol to encrypt the contents of their messages.

S-HTTP

S-HTTP provides the following security checks:

- 1) it authenticates both clients and servers,
- 2) it checks for server certificate revocations,
- 3) it supports; certificate chaining and certificate hierarchies,
- 4) it supports digital signatures that attest to a message's authenticity,
- 5) it allows an application to negotiate the security levels it needs, and
- 6) it provides secured communications through existing corporate firewalls.



Web Browser
HTTP Messages
S-HTTP Seal
SSL
Secure Web
Port 443
Server

Self study:

thin client. Pdf

(Management and Scalability, Access,Performance, Security, and Total Cost of Ownership)

THANK YOU